

CONFINED DIVISORS

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1. REVIEW

This section begins the proof of 4-flips in earnest. The 4-fold case involves some new technical details, which were completely hidden in the 3-fold case. For this reason let me first try to give an outline of the scheme of the proof of the existence of 4-flips.

It suffices to construct the flip in the following situation

- (1) we have a small birational contraction $\#f: \#X \longrightarrow \#Z$ of relative Picard number one,
- (2) $K_{\#X} + X + \#B$ is plt, where X is integral and non-empty,
- (3) $-(K_{\#X} + X + \#B)$ and X are $\#f$ -ample.

This is equivalent to proving that the algebra

$$\#R = R(\#X, K_{\#X} + X + \#B),$$

is finitely generated, which in turn is equivalent to proving that the restricted algebra

$$\text{res}_X \#R,$$

is finitely generated, where the restricted algebra is by definition the image of $\#R$ under the obvious restriction map. The restricted algebra is a graded functional algebra. To any graded functional algebra one can associate a sequence of mobile b-divisors $\mathbf{M}_\bullet$ which satisfy an obvious additivity relation

$$\mathbf{M}_i + \mathbf{M}_j \leq \mathbf{M}_{i+j}.$$

This sequence gives rise to a graded functional algebra $R = R(X, \mathbf{D}_\bullet)$, also known as a pbd algebra. Then the restricted algebra is finitely generated iff R is finitely generated.

Shokurov then isolates two properties of pbd algebras,

- (1) (boundedness) $\mathbf{D}_i \leq \mathbf{D}$, $\mathbf{D}$ fixed,
- (2) (log canonical asymptotic saturation) there is a natural number d such that

$$\text{Mov}^\Gamma j\mathbf{D}_i + \mathbf{A}^\Gamma \leq j\mathbf{D}_j,$$

whenever d divides i and j ,

both shared by restricted algebras, which conjecturally imply that the given algebra is finitely generated:

Conjecture 1.1 (FGA). *Suppose that we are given a pbd algebra $R = R(X, \mathbf{D}_\bullet)$, a morphism $X \rightarrow Z$ and a boundary Δ , where*

- (1) $K_X + \Delta$ is kawamata log terminal,
- (2) $X \rightarrow Z$ is proper, and Z is affine,
- (3) $-(K_X + \Delta)$ is relatively big and nef,
- (4) $\mathbf{D}_\bullet$ is bounded, and
- (5) $\mathbf{D}_\bullet$ is log canonically asymptotically saturated.

Then R is finitely generated.

In fact any pbd algebra which satisfies (1-5) is called a Shokurov algebra.

To prove (1.1) the key point is to prove that the characteristic system

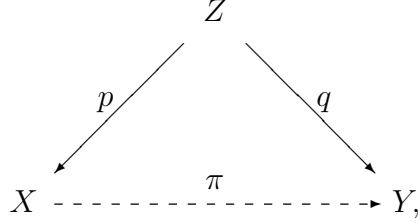
$$\mathbf{D}_i = \frac{\mathbf{M}_i}{i}$$

stabilises. Shokurov is able to prove $(\text{FGA})_{\leq 2}$. Unfortunately this is where the 4-fold case starts to diverge from the 3-fold case, since Shokurov is unable to prove $(\text{FGA})_3$. At the moment, to prove the existence of 4-fold flips, we have to carry around the extra baggage of the ambient 4-fold $\sharp X$.

Nevertheless Shokurov is able to state a general conjecture, the CCS conjecture, whose statement is independent of the existence of the ambient variety X . $(\text{CCS})_{n-1}$ implies $(\text{FGA})_{n-1}$ which in turn implies flips exist in dimension n . Although Shokurov does not prove $(\text{CCS})_3$, the ideas behind trying to prove this conjecture provide the framework for the rest of the proof in the book. In fact to compensate for the fact that he is unable to prove $(\text{CCS})_3$, Shokurov proves that a larger class of pbd algebras on a surface than Shokurov algebras are finitely generated, by virtue of proving a very strong version of $(\text{CCS})_2$. Unfortunately another unpleasant feature of the proof in dimension four is the fact that the birational geometry of a 3-fold is quite a bit more complicated than the birational geometry of a surface. For example, in general there are multiple terminal models, all isomorphic in codimension one.

Definition 1.2. *Let $(X, B = B_X)$ be a log pair, that is $K_X + B$ is $\mathbb{R}$ -Cartier. A **crepant model** of (X, B_X) is a pair (Y, B_Y) , together with a birational map $\pi : X \dashrightarrow Y$ such that for some commutative*

diagram



we have

$$p^*(K_X + B_X) = K_Z + B_Z = q^*(K_Y + B_Y).$$

In particular note that the discrepancy b-divisor is invariant up to a choice of crepant model.

2. DIOPHANTINE APPROXIMATION

In this section we recall some of the arguments that follow subsection 4.3 of [1]. The idea there was to recycle the one dimensional case. The problem in higher dimensions is that $(\mathbf{M}_i)_X$ need not be base point free. The idea is to find an appropriate model Y . In higher dimensions the choice of Y is more problematic and in fact a key part of the proof of the existence of flips. It turns out that a minimal reasonable requirement is that $\mathbf{M}_Y$ is nef; in practice we require even more that $\mathbf{M}_Y$ is semiample. Of course for each i we can find Y_i so that $(\mathbf{M}_i)_{Y_i}$ is base point free. The problem is to find one model that works universally for all i . (In dimension two the choice of this model presents no problems; pick Y to be the unique terminal model of the kawamata log terminal pair (X, B) .)

We first recall the limiting criterion, see (3.7.1) of [1]:

Theorem 2.1. (*Limiting Criterion*) *Let X be a normal variety and let $X \rightarrow Z$ be a proper birational morphism to an affine variety.*

Then the pbd algebra $R = R(X, \mathbf{D}_\bullet)$ is finitely generated iff there is an integer k such that the subsequence $\mathbf{D}_{k\bullet}$ is constant.

Theorem 2.2. *Let (X, B) be a kawamata log terminal pair and let $\pi: X \rightarrow Z$ be a proper birational morphism to the affine variety Z , such that $-(K_X + B)$ is big and nef. Let $R = R(X, \mathbf{D}_\bullet)$ be a Shokurov algebra on X , and let $\mathbf{D} = \lim_i \mathbf{D}_i$.*

Suppose that we may find a variety Y/Z , such that

- (1) $\mathbf{D}_Y$ is semiample
- (2) $\mathbf{D}$ descends to Y .

Then $\mathbf{D}_i$ descends to Y and R is finitely generated.

Proof. We first prove that $\mathbf{D}$ is rational. Since $\mathbf{D} = \bar{D}$, it suffices to prove that D is rational. Suppose not. Then we may apply the Nonvanishing Lemma 4.3.4 of [1], to conclude that there is a positive integer m and an integral divisor M on Y with the following properties

- (i) the linear system $|M|$ is base point free,
- (ii) $mD - M$ is not effective, and
- (iii) for every model $g: W \rightarrow Y$

$$\text{Mov}^\Gamma m\mathbf{D}_W + \mathbf{A}_W^\top \geq g^*M.$$

Since the round up takes values in a discrete topological space (a lattice in a real vector space), it follows that we may find $i > 0$, such that

$$\text{Mov}^\Gamma(m\mathbf{D}_i + \mathbf{A})_Y^\top \geq M.$$

and so pulling back to W , we get

$$\text{Mov}^\Gamma(m\mathbf{D}_i + \mathbf{A})_W^\top \geq g^*M.$$

On the other hand, by asymptotic saturation, we may find a good resolution $g: W \rightarrow Y$, such that

$$\text{Mov}^\Gamma m\mathbf{D}_{iW} + \mathbf{A}_W^\top \leq m\mathbf{D}_{mW} \leq m\mathbf{D}_W.$$

But then

$$g^*M \leq m\mathbf{D}_W = g^*(mD_Y),$$

so that $m\mathbf{D}_Y - M$ is effective, which contradicts (ii).

It follows that $\mathbf{D}$ is rational. Then by a version of the non-vanishing lemma in the rational case, we obtain

$$g^*M \leq m\mathbf{D}_W.$$

As before, by saturation, we may pick g such that

$$\text{Mov}^\Gamma(m\mathbf{D}_W + \mathbf{A})_W^\top \leq m\mathbf{D}_{mW}.$$

It follows that

$$\mathbf{D}_m = \overline{M} = \mathbf{D}.$$

Thus the characteristic sequences stabilises on Y and so by (2.1) we are done. $\square$

Thus there is now a clear strategy to prove $(\text{FGA})_n$. Find sufficient conditions to guarantee the existence of a model satisfying (1) and (2). Note that (1) holds, if B_Y is effective, since then we may apply the base point free Theorem. In practice then it is condition (2) of (2.2) that is the most relevant.

3. DESCENT

In this section we present a simple criteria to ensure that (2) of (2.2) holds.

Definition 3.1. Let $\mathbf{D}$ be a b -divisor. The **obstruction divisor** $\mathbf{O} = \mathbf{O}(X, \mathbf{D})$ for some model X is

$$\mathbf{O} = \overline{\mathbf{D}}_X - \mathbf{D}.$$

Remark 3.2. The obstruction divisor is exceptional over X . It follows that the obstruction divisor only depends on the numerical class of $\mathbf{D}$. Note also that the obstruction divisor is effective provided that $\mathbf{D}$ is b -nef.

Lemma 3.3. Let (X, B) be a log pair. Let $\mathbf{D}_\bullet$ be a sequence of b -divisors, which approaches a limit $\mathbf{D}$. Suppose that we can find a sequence of positive integers $r_1, r_2, \dots$, such that $\lim_{i \rightarrow \infty} r_i = \infty$ and

$$\lceil \mathbf{A}(X, B) - r_i \mathbf{O}_i \rceil \geq 0,$$

for every exceptional divisor over X , where $\mathbf{O}_i = \mathbf{O}(X, \mathbf{D}_i)$.

Then $\mathbf{D}$ descends to X .

Proof. This is easy. Indeed, let $g: W \rightarrow Y$ be any model over Y , $D_i = (\mathbf{D}_i)_Y$, $\mathbf{O}_i = \mathbf{O}(Y, \mathbf{D}_i)$, $D = \mathbf{D}_Y$ and $\mathbf{O} = \mathbf{O}(Y, \mathbf{D})$. We have

$$\lceil \mathbf{A}(X, B)_W \rceil - \lfloor r_i \mathbf{O}_i \rfloor \geq 0.$$

Hence

$$\begin{aligned} \mathbf{O}_W &= g^*D - \mathbf{D}_W \\ &= \lim_i (g^*D_i - (\mathbf{D}_i)_W) \\ &= \lim_i (\mathbf{O}_i)_W \\ &= \lim_i \frac{1}{r_i} \lfloor r_i (\mathbf{O}_i)_W \rfloor \\ &\leq \lim_i \frac{1}{r_i} \lceil \mathbf{A}_W \rceil = 0. \end{aligned}$$

On the other hand $\mathbf{O}_W \geq 0$. □

Combining (2.2) and (3.3), to prove that FGA, it suffices to find a model Y on which every nef divisor is semiample, satisfying the hypothesis of (3.3). Shokurov's clever idea is to drop any reference to the sequence of b -divisors $\mathbf{D}_\bullet$ and work instead with general log canonically saturated b -divisors.

4. CONFINED B-DIVISORS

Definition 4.1. We will say that the log pair (X, Δ) is **canonical** if every exceptional divisor has log discrepancy greater than one and if the coefficient of every component of Δ is less than one.

Definition 4.2. Let X be a variety and let B be a divisor such that $K_X + B$ is terminal. Let $\mathfrak{F}$ be a set of divisors. We say that $\mathfrak{F}$ is **canonically confined by c** if for every $G \in \mathfrak{F}$, $K_X + B + cG$ is canonical.

We say that $\mathfrak{F}$ is **canonically confined by c up to linear equivalence** if for every $G' \in \mathfrak{F}$ there exists $G \in |G'|$ such that $K_X + B + cG$ is canonical.

There are similar definitions, where canonical is replaced by log canonical. There are similar notions for b-divisors. But note the following subtlety:

Example 4.3. Note that linear equivalence of b-divisors makes sense. We define $|\mathbf{D}|$ to be the linear system of all effective b-divisors linearly equivalent to $\mathbf{D}$. We have

$$|\mathbf{D}|_X \subset |\mathbf{D}_X|,$$

and in general the inequality is strict. For example let L be a line in $X = \mathbb{P}^2$ and let $\mathbf{D}$ be the b-divisor given by the single prime divisor L . Then $\mathbf{D}_X = L$ so that $|\mathbf{D}_X|$ is the complete linear system of lines in $\mathbb{P}^2$. However $|\mathbf{D}|_X = \{L\}$. A more interesting example arises if we take $Y \rightarrow X$ the blow up of $\mathbb{P}^2$ at a point p , M the strict transform of L and we set $\mathbf{M} = \overline{M}$. In this case $|\mathbf{M}_X|$ is again the complete linear system of lines in $\mathbb{P}^2$ and $|\mathbf{M}|_X$ is the linear system of lines through p . The whole point is that $\mathbf{D}$ is not exceptionally saturated. Indeed

$$\text{Mov}(\mathbf{M}_Y + E) = \text{Mov}(M + E) = M + E > M = \mathbf{D}_Y.$$

Lemma 4.4. Suppose that we have a sequence of b-divisors $\mathbf{D}_\bullet$ on $(X/Z, B)$. Suppose that the set $\mathfrak{F} = \{\mathbf{M}_i\}$ is log canonically confined by c up to linear equivalence on a crepant model (Y, B_Y) .

Then (2) of (2.2) holds.

Proof. By assumption, for every i , we may find $\mathbf{G}_i \in |\mathbf{M}_i|$ such that $(Y, B + c\mathbf{G}_i)$ is log canonical, that is

$$\lceil \mathbf{A}(X, B + c\mathbf{G}_i) \rceil \geq 0,$$

where G_i is the trace of $\mathbf{G}_i$ on Y . Hence

$$\begin{aligned}\lceil \mathbf{A}(X, B) - ci\mathbf{O}(\mathbf{D}_i) \rceil &= \lceil \mathbf{A}(X, B) - c\mathbf{O}(\mathbf{M}_i) \rceil \\ &= \lceil \mathbf{A}(Y, B_Y) - c\mathbf{O}(\mathbf{G}_i) \rceil \\ &= \lceil c\mathbf{G}_i + \mathbf{A}(Y, B_Y) - c\overline{\mathbf{G}}_i \rceil \\ &= \lceil c\mathbf{G}_i + \mathbf{A}(Y, B_Y + cG_i) \rceil \geq 0.\end{aligned}$$

Now apply (3.3). $\square$

Thus we would like to find sufficient conditions such that a set of b-divisors is log canonically confined up to linear equivalence.

Lemma 4.5. *Let $\mathfrak{F}$ be a set of divisors.*

- (1) *If $\mathfrak{F}$ is canonically confined then it is log canonically confined. The converse holds provided that $K_X + B$ is terminal.*
- (2) *$\mathfrak{F}$ is log canonically confined iff the log canonical threshold of the elements of $\mathfrak{F}$ is bounded away from zero.*
- (3) *Suppose that X is smooth. Then $\mathfrak{F}$ is canonically confined iff the multiplicity of the elements of $\mathfrak{F}$ is bounded.*
- (4) *If $\mathfrak{F}$ forms a bounded set (in the sense of moduli) then $\mathfrak{F}$ is canonically confined.*
- (5) *If every element of $\mathfrak{F}$ is base point free then $\mathfrak{F}$ is canonically confined up to linear equivalence.*

There are similar notions for b-divisors and linear equivalence, whenever appropriate. In particular (5) holds for b-divisors associated to base point free divisors.

(4) and (5) in conjunction present a strategy to prove that a set $\mathfrak{F}$ of b-divisors is log canonically confined up to linear equivalence. Prove that the subset of b-divisors that are not base point free is bounded. In practice it turns out that, at least for a surface, it is easier to prove this on a crepant terminal pair. For this reason Shokurov prefers to work with the notion of canonically confined.

5. THE STANDARD CONJECTURES

In this section we state a version of the CCS conjecture, and we develop some general techniques to attack this conjecture. Unfortunately it is not clear how to formulate the CCS conjecture. Clearly we want a conjecture that is strong enough to imply the FGA conjecture. On the other hand, we want a statement which does indeed hold, and which someone might hopefully prove in the future.

With this said, in the interests of pedagogy, we give the easiest version of the CCS conjecture to state, and we have limited ourselves to

stating only one version. Anyone with a serious interest in this topic is well advised to consult the paper of Shokurov [3], which by way of contrast states many different conjectures, which are all variations on the following theme:

Conjecture 5.1 (CCS). *Suppose that $K_X + B$ is kawamata log terminal and $-(K_X + B)$ is nef and big over Z . Then there is a constant $c > 0$ and a bounded family*

$$\{ (Y_t, T_t) \mid t \in U \},$$

where (Y_t, B_t) is a crepant $\mathbb{Q}$ -factorial model of X , there are contraction morphisms $\psi_t: Y_t \rightarrow T_t$, $\phi_t: T_t \rightarrow Z$, where the relative cone of nef divisors of ϕ_t is spanned by a finite number of semiample divisors, such that for every mobile and $\mathbf{A} = \mathbf{A}(X, B)$ -saturated b -divisor $\mathbf{M}$, we may find $t \in U$ with the following properties

- (1) *We may find $G \in |\mathbf{M}|_{Y_t}$ such that $(Y, B_Y + cG)$ is log canonical.*
- (2) *$\mathbf{M}_Y = \psi_t^* M$ where M is a big and nef divisor on T_t .*

Moreover we may choose U to be finite if further we fix a contraction Y/T and we restrict to those $\mathbf{M}$ which define this contraction.

Remark 5.2. *Note that the condition that the relative cone of nef divisors of ϕ_t is spanned by a finite number of semiample divisors is quite a stringent condition. In practice, the most natural way to satisfy this condition is to require that there is a divisor Θ on T such that $-(K_T + \Theta)$ is nef and big over Z .*

Theorem 5.3. *$(CCS)_n$ implies $(FGA)_n$.*

Proof. We are given a convex bounded sequence $\mathbf{M}_\bullet$ of b -free log canonically asymptotically saturated divisors and it suffices to prove the sequence $\mathbf{D}_\bullet$ stabilises. By convexity, possibly up to truncation, the divisors $\mathbf{M}_i$ define the same contraction. Thus by $(CCS)_n$ we may assume that there is a fixed pair $(Y/T/Z, B_Y)$ which is desirable for infinitely many of the b -divisors $\mathbf{M}_i$. As $\mathbf{M}_i$ is pulled back from T , then so is $\mathbf{D}_Y$ and as every nef divisor on T is semiample, $\mathbf{D}_Y$ is semiample. The rest is automatic from the definition of a pair and a prediction. $\square$

Example 5.4. *Let $f: X \rightarrow Z$ be a small birational contraction of threefolds of relative Picard number one. Assume, for example, that both X and the exceptional locus C is smooth and that there is a surface D which is f -negative.*

Consider the constant sequence of b -divisors, $\mathbf{D}_i = \overline{D}$. Then the sequence of b -divisors $\mathbf{M}_i$ is not canonically confined. Indeed suppose that it were. First note that canonically confined and log canonically confined are equivalent. Thus there would be a constant $c > 0$ and

divisors $S_i \sim iD$ such that $K_X + c_i D_i$ is maximally log canonical at the generic point of C , where $c_i \geq c$.

But then by codimension two subadjunction

$$(K_X + c_i S_i)|_C = K_C + G_i$$

where G_i is effective. On the other hand, the degree of the left hand side goes to $-\infty$ whilst the right hand side is bounded from below, a contradiction.

On the other hand if we can find the opposite $f^+: X^+ \rightarrow Z$ of f , then the existence of these divisors on X^+ is trivial, since D^+ is ample. This shows that we need at least nef in the CCS conjecture. If f is a flopping contraction then this also shows that the family of desirable pairs above does not always consist of a single point.

We present a general strategy to prove the CCS conjecture, and show how this strategy is relatively straightforward to implement in the case of surfaces.

Suppose that we are given a log pair $K_X + B$ and an integral b -free b -divisor $\mathbf{M}$. Our aim is to control the base locus of the linear system $|\mathbf{M}|_X$. To this end, we pick a general element $S \in |\mathbf{M}|_X$. We work on some high model $\pi: Y \rightarrow X$ over X . In particular we assume that Y is smooth and $|\mathbf{M}|_Y$ is free. Let T be the divisor in $|\mathbf{M}|_Y$ lying over S .

Clearly the base locus of $|\mathbf{M}|_X$ is contained in S . As usual, we restrict to T and analyse the base locus on T . However, instead of restricting $|\mathbf{M}|_Y$ to T , we consider the restriction L of the line bundle $\mathcal{O}_Y(\mathbf{M} + \lceil \mathbf{A}_Y \rceil)$. In this case the base locus of $|\mathbf{M}|_X$ is related to the base locus of $|L|$ in a much more indirect fashion.

Lemma 5.5. *Let $K_X + B$ be a log pair over Z and let $\mathbf{M}$ be an integral b -free b -divisor. Pick a general element $S \in |\mathbf{M}|$.*

Then we may find a sufficiently high birational model $\pi: Y \rightarrow X$, with the following properties

- (a) $|\mathbf{M}|_Y$ is free.
- (b) Y union the support of the inverse image of B and Z is a divisor with normal crossings.
- (c) The locus $V \subset X$ where π is not an isomorphism is the union of the locus where the pair (X, B) does not have normal crossings and the base locus of $|\mathbf{M}|_X$.

Further, if Y is any model which satisfies (a-c), then Y has the following additional property:

- (d) If $\mathbf{M}$ is $\mathbf{A}$ -saturated, then $\mathbf{M}_Y$ is $\mathbf{A}(Y, B_Y)$ -saturated over Z .

Proof. Let $\psi: W \rightarrow X$ be a log resolution of the pair $(X, B + S)$. Then ψ is certainly an isomorphism outside of the locus where the pair $(X, B + S)$ does not have normal crossings. As $\mathbf{M}$ is b -free, S is certainly reduced, and the locus where $(X, B + S)$ does not have normal crossings, is contained in the locus where the pair (X, B) does not have normal crossings, union the base locus of $|\mathbf{M}|_X$. Blowing up the base locus of $|\mathbf{M}|_W$, we may assume that (a-c) hold.

By (3.3.8) of [1], (d) holds. $\square$

Definition 5.6. Let $K_X + B$ be a log pair over Z and let $\mathbf{M}$ be an integral b -free b -divisor. Pick a general element $S \in |\mathbf{M}|_X$ and pick a model $\pi: Y \rightarrow X$ satisfying (a-c) of (5.5). Let V be the locus where π is not an isomorphism and F the inverse image of V . Let T be the divisor in $|\mathbf{M}|_Y$ lying over S , and let L be the restriction to T of the line bundle $\mathcal{O}_Y(\mathbf{M} + \lceil \mathbf{A}_Y \rceil)$.

We say that the pair $(L, S \in |\mathbf{M}|_X)$ is g -free if

- (1) The base locus of $|L|$ is contained in F .
- (2) Given any point $p \in V$, we may find $S \in |\mathbf{M}|_X$ such that the base locus of $|L|$ does not intersect the fibre $\pi^{-1}(p)$.

Here the g of g -free stands for generic. Note that even though g -free is a property of both L and the choice of S , we will abuse notation and often drop the reference to the choice of S .

Remark 5.7. Note that if the singular locus of X is at most zero dimensional, then $|L|$ is g -free iff $|L|$ is free, since in this case S is disjoint from the singular locus of X . In particular this holds if X has dimension at most two, or if X is a terminal threefold. In general however, S will always intersect the singular locus of V and the inverse image of this locus is always contained in the base locus of L .

Proposition 5.8. Suppose that $-(K_X + B)$ is nef and big over Z , where $K_X + B$ is terminal. Let $\mathbf{M}$ be an integral b -free b -divisor which is $\mathbf{A} = \mathbf{A}(X, B)$ -saturated over Z . Let $S \in |\mathbf{M}|_X$ be a general element. Pick a birational model $\pi: Y \rightarrow X$ satisfying (a-c) of (5.5).

Let $T \in |\mathbf{M}|_Y$ be the general element corresponding to S , and let L be the restriction of the line bundle $\mathcal{O}_Y(\mathbf{M}_Y + \lceil \mathbf{A}_Y \rceil)$ to T .

- (1) Let B be a component of the base locus of $|\mathbf{M}|_X$ and let E be an exceptional divisor, with centre B . Then every component E' of $E \cap T$ is a component of the base locus of $|L|$.
- (2) $|\mathbf{M}|_X$ is free iff L is g -free.
- (3) The line bundle L has the form

$$c_1(L) = K_T + \Delta + D,$$

where Δ has normal crossings, every coefficient lies between zero and one, and $D = (-\pi^*(K_X + \Delta))|_T$ is the pullback of a nef and big divisor from S .

Proof. We first prove (1). Now, as $\mathbf{M}_Y$ is $\mathbf{A}_Y$ saturated, it follows that

$$\text{Mov } |\mathbf{M}_Y + \lceil \mathbf{A}_Y \rceil| \leq |\mathbf{M}_Y|.$$

By assumption $|\mathbf{M}|_Y$ is a free linear system. As the pair (X, B) is kawamata log terminal, $\lceil \mathbf{A}_Y \rceil$ is effective, and as the pair (X, B) is terminal, the support of $\lceil \mathbf{A}_Y \rceil$ is precisely the exceptional locus. Thus every component of the exceptional locus is a component of the base locus of the linear system $|\mathbf{M}_Y + \lceil \mathbf{A}_Y \rceil|$.

Let B be a component of the base locus of the linear system $|\mathbf{M}|_X$, and let E be an exceptional divisor on Y with centre B . By what we just said, E is a component of the linear system $|\mathbf{M}_Y + \lceil \mathbf{A}_Y \rceil|$. Let E' be any component of $E \cap T$. Consider the restriction exact sequence,

$$0 \longrightarrow \mathcal{O}_Y(\lceil \mathbf{A}_Y \rceil) \longrightarrow \mathcal{O}_Y(T + \lceil \mathbf{A}_Y \rceil) \longrightarrow L \longrightarrow 0.$$

Taking global sections, we get a restriction map on linear systems

$$|T + \lceil \mathbf{A}_Y \rceil| \longrightarrow |L|.$$

This map is surjective as

$$H^1(Y, \lceil \mathbf{A}_Y \rceil) = H^1(Y, K_Y + \lceil -f^*(K_X + B) \rceil) = 0,$$

by Kawamata-Viehweg vanishing. Suppose that E' is not a component of the base locus of $|L|$. Then we could find a section of L which does not vanish at any point of E' . By surjectivity, we could lift this section to Y , and it would follow that E is not a component of the base locus of $|T + \lceil \mathbf{A}_Y \rceil|$, a contradiction. Thus E' is a component of the base locus of $|T + \lceil \mathbf{A}_Y \rceil|$. Hence (1).

We turn to the proof of (2). By assumption $|\mathbf{M}|_Y$ is a free linear system. Since $\mathbf{A}_Y$ is exceptional and $\lceil \mathbf{A}_Y \rceil$ is effective, it follows that the only possibly base locus of the linear system $|\mathbf{M}_Y + \lceil \mathbf{A}_Y \rceil|$ is supported on the exceptional locus of π . We have a commutative diagram,

$$\begin{array}{ccccccc} T & & \subset & Y & & \supset & F \\ \downarrow f & & & \downarrow \pi & & & \downarrow \\ S & & \subset & X & & \supset & V. \end{array}$$

Suppose that $|\mathbf{M}|_X$ is free. Then the only locus where the map f is not an isomorphism, is supported in $V \cap S$. As $|\mathbf{M}|_X$ is free, given any

point V , we may always pick an S that does not contain this point, so that T will not even intersect the fibre over p .

Now suppose that the base locus of the linear system $|L|$ is supported in V , and that we may avoid the fibre over any point p of V . Suppose that $|M|_X$ is not free. Pick a component B of the base locus, and a point $p \in B$. Let q be point in the fibre over p . Then there must be an exceptional divisor E , containing q , with centre B which intersects T , since the linear system $|M|_Y$ is free. But then any component E' of $E \cap T$ is a component of the base locus of $|L|$, by (1), and one of these components contains q . But then the whole fibre over p is in the base locus of $|L|$, which contradicts our assumption that L is g-free. This proves (2).

Finally we prove (3). We have

$$\begin{aligned} c_1(L) &= (T + \mathbf{A}_Y)|_T \\ &= (T + K_Y + {}^\top(-\pi^*(K_X + B))^\top)|_T \\ &= K_T + {}^\top D^\top \\ &= K_T + \Delta + D, \end{aligned}$$

where we applied adjunction to get from line two to line three, D is the restriction of $-(K_Y + B)$ to T and Δ is defined as the difference ${}^\top D^\top - D$. Now $-(K_X + B)$ is big and nef over Z , so that D is certainly nef over Z . However as T is the general element of a base point free linear system, it follows that D is also big over Z . Clearly the coefficients of Δ lie between zero and one. On the other hand, the support of Δ is contained in the support of the strict transform of B and the exceptional locus. As this has normal crossings in X and T is the general element of a base point free linear system, it follows that Δ has normal crossings. Hence (3). $\square$

Clearly, given (5.8), we should study carefully the base locus of adjoint line bundles. In general, since we are only assuming that D is big and nef, and we make no other assumption about the positivity of L , and since we want to prove that $|L|$ is close to being free, rather than some multiple of L , this is rather a tall order. Fortunately, however, to prove $(CCS)_2$ we only have to consider adjoint line bundles on a curve, which are particularly easy.

6. THE CASE OF SURFACES

Theorem 6.1. $(CCS)_{\leq 2}$ holds.

We first note that $(\text{CCS})_1$ is easy; indeed if X is a curve, there is only one model and so $|\mathbf{M}|_X$ is, by hypothesis, free. In this case, any $c < 1$ will do, since by (4.1) we only need to ensure that every coefficient of $K_X + \Delta + cS$ is less than one.

Lemma 6.2 (Riemann's Theorem). *Let C be a smooth curve, and let L be a line bundle of the form $K_C + \Delta + D$, where D is big and nef and every coefficient of Δ lies between zero and one.*

If $|L|$ is not base point free, then C is complete and $D + \Delta = p$. In particular the base locus is the single point p , and $\Delta = \lambda p$, for some $0 < \lambda \leq 1$.

Proof. Well known. □

Using (6.2) we are able to give, in the case of surfaces, a sharp classification of b-free, saturated divisors which are not free. (6.1) will then be an easy consequence of this classification.

Definition 6.3. *Let X be a normal variety and let $\mathbf{M}$ be a b-divisor. The **Kodaira dimension** of $\mathbf{M}$ is equal to the maximal dimension of the image of X , under the linear systems $|\mathbf{M}|_X$.*

Proposition 6.4. *Suppose that $-(K_X + B)$ is nef and big over Z , where $K_X + B$ is terminal and X is a projective surface over an affine variety Z . Let $\mathbf{M}$ be an integral b-free b-divisor which is $\mathbf{A} = \mathbf{A}(X, B)$ -saturated over Z .*

Suppose that $|\mathbf{M}|_X$ is not free. Then Z is a point and for every component C of any element $|\mathbf{M}|_X$, $-(K_X + B) \cdot C \leq 1$. Moreover either,

- (i) $\mathbf{M}$ is big and $-(K_X + B) \cdot \mathbf{M}_X \leq 1$, or
- (ii) *There is a point $x \in X$ and a birational morphism $\psi: W \rightarrow X$, where the linear system $|\mathbf{M}|_W$ defines a morphism $f: W \rightarrow T$ to a curve whose general fibres are elliptic curves. Moreover x is the only point in X where ψ is not an isomorphism and the general fibre of f meets exactly one ψ -exceptional divisor E . Further the log discrepancy of E is at most two with respect to $K_X + B$.*

Proof. Since $\mathbf{M}$ is b-free, it follows that $|\mathbf{M}|_X$ is a mobile linear system and that the general element of $|\mathbf{M}|_X$ is reduced. The proof breaks into cases, depending on the Kodaira dimension of $\mathbf{M}$. If the Kodaira dimension is zero, then in fact $\mathbf{M} = 0$ and the base locus is certainly empty.

Otherwise we are going to apply (5.8). Pick a general element $S \in |\mathbf{M}|_X$ and pick a model $\pi: Y \rightarrow X$ satisfying (a-c) of (5.5). Let

$T \in |\mathbf{M}|_Y$ be the curve lying over S and let L be the restriction of $\mathbf{M}_T + \lceil \mathbf{A}_T \rceil$ to T .

By (b) of (5.8), it suffices to consider the case when $|L|$ is not free. Let C be a component of T . Note that by (6.2), C must be complete, and L has the form $K_T + \Delta + D$, where every coefficient of Δ lies between zero and one and D is big and nef.

Now the only complete curves are fibres of $X \rightarrow Z$. Suppose that Z is not a point. If Z is a surface, then there are only finitely many complete curves, which contradicts the fact that $|\mathbf{M}|_X$ has no base divisors. Thus Z is a curve and C is a fibre of $X \rightarrow Z$. Since no two fibres intersect, the fact that $\mathbf{M}$ is b -free implies that $|\mathbf{M}|_X$ is free, a contradiction.

It follows that Z must be a point. We apply (6.2) to C . It follows that $D = \lambda p$, for some $0 < \lambda \leq 1$, and

$$\begin{aligned} -(K_Y + B_Y) \cdot C &= \deg D \\ &= \lambda \\ &\leq 1. \end{aligned}$$

If the Kodaira dimension is two, so that $\mathbf{M}$ is big, then $T = C$ is irreducible. This gives (i).

The only remaining case is when the Kodaira dimension of $\mathbf{M}$ is one. It follows that the restriction to C of $\mathbf{M}$ has no movable part. Let g be the genus of C , so that L has degree equal to $2g - 1$. Now $g \neq 0$, as L is nef. Note that, by saturation, the movable part of $|L|$ is equal to the restriction of the linear system $|M|_X$ to C . Thus $|L|$ has empty movable part, and so $g < 2$. Hence $g = 1$ and C is a smooth elliptic curve.

Let $x \in X$ be the point, corresponding to the unique base point p of C . Then x is the unique base point of the linear system $|\mathbf{M}|_X$. Thus we may find a birational morphism $\psi: W \rightarrow X$ such that the linear system $|\mathbf{M}|_W$ is free, where x is the only point where ψ is not an isomorphism. The fibres of ψ are precisely the curves C , so that the fibres have genus one. Since $\lceil \mathbf{A}_Y \rceil$ restricted to C has support a single point, it follows that the general fibre of ψ meets exactly one exceptional divisor E , and since $\lambda \leq 1$, the intersection number is one. $\square$

Note that in case (ii), we have to prove some sort of boundedness. We present two different ways of doing this. Here is one of them. Given any valuation ν of X , consider the sequence of blow ups obtained by iteratively blowing up the centre of ν . We will call this sequence of

blow ups the Zariski tower. We say that ν is an algebraic valuation, if this sequence stabilises, in the sense that ν corresponds to a divisor. We call the first model on which the centre of ν is a divisor, the top of the tower. The number of blows ups to get X to the top is called the length of the tower.

Definition 6.5. *Let X be a variety and let $\mathfrak{V}$ be a set of algebraic valuations on X . We say that $\mathfrak{V}$ forms a **bounded family**, if there is a bounded family $Y \longrightarrow B$ of varieties, such that for every $\nu \in \mathfrak{V}$, there is a $t \in B$ such that Y_t is birational to X and ν is a divisorial valuation on Y_t .*

Remark 6.6. *Note that if $\mathfrak{V}$ is a bounded set of valuations, then we may further assume that for each t , there is a birational morphism $Y_t \longrightarrow X$.*

Lemma 6.7. *Fix a log terminal pair (X, B) , where X is a surface. Fix a real number b .*

Then the set of valuations of log discrepancy at most b is bounded.

Proof. Since the set of all surfaces that can be obtained from X by a fixed number of blow ups forms a bounded family, it suffices to prove that there is a uniform bound on the length of a valuation ν of log discrepancy at most b .

Passing to a log resolution, we might as well assume that the pair (X, B) has normal crossings. Pick m such that $m(K_X + B)$ is Cartier. It follows that if ν is an algebraic valuation of log discrepancy a , then ma is an integer. In particular there are only finitely many possible values a of the log discrepancy less than b .

It suffices then, to prove that for any tower of blow ups, the log discrepancy of each successive exceptional divisor goes up strictly. So suppose that we are at some step Y of the Zariski tower and that the centre of ν on Y is q . Then at most two components of $C = B_Y$ pass through q , with coefficients $x \leq y$ say. In this case, by direct calculation, the exceptional divisor has log discrepancy

$$2 - x - y > 1 - x \geq 1 - y,$$

and we are done by induction. $\square$

Remark 6.8. *Note that the corresponding result in higher dimensions fails. The problem is that we have no control over the degrees of the centres of any valuation of bounded log discrepancy, so that knowing that the length of the Zariski tower is bounded is of no use.*

Proof of (6.1). Let X be a surface over Z and let (Y, B_Y) be a terminal model. We are looking for a bounded family of pairs $W/T/Z$, such

that $\mathbf{M}_W$ is nef and pulled back from T , every nef divisor on T is semiample, and there is a constant c such that there is an $S \in |\mathbf{M}|_W$ with $K_W + B_W + cS$ canonical. We will choose models W over Y .

Note first that $\mathbf{M}_W$ is automatically nef on any model W , as the base locus is a finite set of points. Suppose first that $|\mathbf{M}|_Y$ is base point free or big. In this case the pair $(Y/Y/Z, B_Y; \mathfrak{F})$ is desirable for $\mathbf{M}$. Indeed, since $-(K_Y + B_Y)$ is nef and big over Z , it is automatic that every nef divisor is semiample, by the base point free Theorem. If $|\mathbf{M}|_Y$ is free, then we may take $c = 1$. If $\mathbf{M}$ is big, but not free, then by (6.4) Z is a point and

$$-(K_Y + B_Y) \cdot \mathbf{M}_Y \leq 1.$$

As $|\mathbf{M}|_Y$ has no base components, it follows that $S \in |\mathbf{M}|_Y$ forms a bounded family and the existence of c is then clear.

Otherwise, by (6.4), it follows that $\mathbf{M}_Y$ defines a pencil, and there is a model $\psi: W \rightarrow Y$ and a morphism $f: W \rightarrow T$ down to a curve T , such that the only exceptional divisor E which is horizontal for f has log discrepancy at most two. As $|\mathbf{M}|_W$ is base point free, we may take any $c < 1$. As E dominates T , $T \simeq \mathbb{P}^1$ and the condition that every nef divisor on T is semiample is trivial.

It suffices to prove that we may take a bounded family of such surfaces W . We prove this in two different ways. One way is to apply (6.7). It follows that the valuations corresponding to E form a bounded family, and from there it is easy to see that the set of all such surfaces W forms a bounded family. On the other hand, we could appeal to the fact that $-(K_Y + B_Y) \cdot C$ is at most one, to conclude that C belongs to a bounded family. It follows that we may resolve the basis locus of the linear system associated to C using a bounded family. But this also then resolves the basis locus of $|\mathbf{M}|_X$ and so W belongs to a bounded family. $\square$

Example 6.9. *Let S be a del Pezzo surface of degree one, that is S is $\mathbb{P}^2$ blown up at eight points. Then $|-K_S|$ defines a pencil with a unique base point p . Let $T \rightarrow S$ blow up this base point, let C be a generic element of this linear system on T and let $\mathbf{D} = \overline{C}$. Then the set $\{\mathbf{M}_\bullet\}$ is not (log) canonically confined on S , even though each $\mathbf{M}_i$ is $\mathbf{A}(S, 0)$ -saturated. Indeed the linear system $|\mathbf{M}_i|_S$ has p as a base point of multiplicity i . So we cannot work only on the terminal model, even in the case of surfaces.*

It is easy to generalise this example. Consider a del Pezzo S of degree d , so that we blow up $k = 9 - d$ points of $\mathbb{P}^2$. Pick a pencil of cubics in the plane, spanned by two cubics, which pass through the k points we blow up, which share a common point and $d - 1$ infinitely points lying

over this point. If we blow up these infinitely near points, $\psi: T \longrightarrow S$, then our pencil defines a morphism $T \longrightarrow \mathbb{P}^1$, whose fibres are the strict transforms of the cubics in the plane. Let C be a general fibre and set $\mathbf{M} = m\overline{C}$, where m is any positive integer, considered as a b -divisor on X , and let E be the unique exceptional divisor which intersects C .

If B_1 denotes the strict transform on X of any of the pencil of plane cubics, then $B_1 \in |-K_X|$. Let $B = aB_1$, for some $0 < a < 1$, so that $-K_X - B = -(1-a)K_X$ is ample, as X is a del Pezzo. We want to choose a so that $\mathbf{M}$ is $\mathbf{A}(X, B)$ -saturated. Once again it suffices to check that the log discrepancy of the valuation corresponding to the last infinitely near point, which corresponds to the unique exceptional divisor E which intersects C , is at most 2. But this log discrepancy is easily seen to be $1 + d(1-a)$ (see (??) of [2]), so that it suffices to pick $a < 1/d$.

Note that this is in contrast to a claim made in [3], where it is claimed that we always only need one blow up, so that $k = 1$. It is however still true that we have a lot of control in this situation, and it is possible to make a complete classification, see §2 of [2].

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